

Intelligent Smart Microgrid Energy Management System using Machine Learning

B.Tech Final Year Project - Electrical and Electronics Engineering

ABSTRACT

The integration of renewable energy sources (RES) into microgrids has created unprecedented challenges in energy forecasting, power management, and operational optimization due to the inherent intermittency and stochasticity of solar and wind power generation. This project proposes an Intelligent Smart Microgrid Energy Management System leveraging advanced machine learning techniques to address these challenges. The system integrates Long Short-Term Memory (LSTM) neural networks for accurate time-series forecasting of renewable energy generation and load demand, coupled with deep reinforcement learning (DRL) algorithms for real-time energy dispatch optimization.

The proposed framework encompasses multiple components: (1) predictive models using LSTM and Support Vector Regression (SVR) for solar PV and wind power forecasting with enhanced accuracy metrics, (2) energy storage optimization using battery management systems with state-of-charge prediction, (3) real-time demand response management through intelligent load scheduling, and (4) cost optimization while maximizing renewable energy utilization. The system will be implemented using MATLAB and Simulink, providing a comprehensive simulation environment for microgrid operations under various scenarios including grid-connected and islanded modes.

Expected outcomes include achieving renewable energy forecasting accuracy with RMSE < 2.0 for solar PV and < 2.5 for wind power, reducing peak grid imports by 15-20%, improving energy self-sufficiency to 85-90%, and demonstrating a 20-25% reduction in operational costs compared to conventional rule-based systems. This research contributes to the advancement of intelligent energy management systems crucial for the transition towards sustainable and resilient power infrastructure.

PROBLEM STATEMENT

Background

Modern power systems are undergoing a fundamental transformation with the rapid integration of distributed energy resources (DERs), particularly renewable energy sources like solar and wind. Microgrids have emerged as a promising solution to enhance energy reliability, reduce transmission losses, and facilitate higher penetration of renewables. However, several critical challenges impede optimal microgrid operation:

Key Problems Identified

1. Renewable Energy Intermittency and Uncertainty

- Solar and wind power generation exhibit high variability due to weather conditions

- Traditional forecasting methods (ARIMA, linear regression) fail to capture complex nonlinear patterns
- Forecast errors lead to energy imbalances, requiring expensive backup power or causing load shedding
- Lack of accurate short-term (15-min to 1-hour ahead) and day-ahead forecasting capabilities

2. Complex Energy Management and Optimization

- Real-time balancing between variable generation, fluctuating demand, and storage systems
- Traditional optimization methods (MILP, dynamic programming) are computationally intensive for real-time decisions
- Multiple conflicting objectives: cost minimization, emission reduction, reliability maximization
- Difficulty in handling multi-time-scale operations (seconds to days)

3. Energy Storage Management Challenges

- Inefficient battery charging/discharging strategies leading to premature degradation
- Inability to predict optimal state-of-charge (SoC) trajectories
- Peak load management without proper forecasting causes battery violations
- Lack of coordination between different storage technologies (batteries, supercapacitors)

4. Grid Integration and Stability Issues

- Voltage and frequency fluctuations due to intermittent RES
- Challenges in seamless transition between grid-connected and islanded modes
- Power quality degradation affecting sensitive loads
- Limited flexibility in demand response programs

5. Economic and Operational Inefficiencies

- Suboptimal energy dispatch leading to higher operational costs
- Excessive dependence on expensive grid electricity during peak hours
- Renewable energy curtailment due to poor scheduling
- Inability to participate effectively in peer-to-peer (P2P) energy trading

Research Gap

While recent studies have explored machine learning for individual aspects (forecasting OR optimization), there is a critical need for an **integrated intelligent system** that combines:

- Multi-algorithm forecasting (LSTM + SVR + Random Forest) for enhanced accuracy
- Deep reinforcement learning for adaptive, real-time decision-making
- Comprehensive energy storage management with predictive control
- Validation through realistic MATLAB/Simulink simulations

Project Objective

Develop and validate an Intelligent Smart Microgrid Energy Management System using machine learning that addresses the above challenges through integrated forecasting, optimization, and control strategies, demonstrated via comprehensive MATLAB/Simulink simulations.

LITERATURE REVIEW

1. Machine Learning for Renewable Energy Forecasting

LSTM and Neural Network Approaches:

Singh et al. (2024) demonstrated Support Vector Regression (SVR) for power forecasting in grid-connected microgrids, achieving MSE of 2.002 for solar PV and 3.059 for wind power, with RMSE of 1.415 and 1.749 respectively. The model leveraged historical energy production data, weather patterns, and dynamic grid conditions, significantly outperforming traditional linear regression models.

Eyimaya et al. (2025) employed LSTM networks for load forecasting in DC microgrid optimization, comparing with Multi-Layer Perceptron (MLP) networks. The LSTM model demonstrated superior capability in capturing both short-term fluctuations and long-term dependencies in sequential data, improving system responsiveness and efficiency for demand-side management.

Silva-Rodriguez et al. (2024) presented LSTM-based net load forecasting specifically for wind and solar power-equipped microgrids, processing temporal sequences efficiently to handle fluctuations in both demand and renewable generation. The model utilized Typical Meteorological Year (TMY) data for training and validation.

Random Forest and Ensemble Methods:

Recent work by researchers (2025) on intelligent energy management using Random Forest regressor demonstrated 18% reduction in peak grid imports and 11% reduction in daily imported energy through accurate forecasting combined with rule-based storage-dispatch strategies. The Random Forest model proved particularly effective for energy prediction when compared with LSTM for load forecasting.

2. Deep Reinforcement Learning for Energy Management

Recent Advancements (2024-2025):

SmartGrid AI Platform (2025) integrated Deep Reinforcement Learning (DRL) and neural networks for optimizing energy consumption, demand prediction, and peer-to-peer (P2P) energy trading. The platform achieved:

- 23% reduction in energy costs
- 40% decrease in grid dependency
- 85% renewable energy utilization rate
- 18% of electricity consumption handled through P2P trading
- Vehicle-to-home (V2H) technology supplying 15% of household demand during peak hours

Algorithm-Specific Studies:

Proximal Policy Optimization (PPO): Research from 2025 showed PPO agents were up to 9.2% more effective in reducing annual costs and carbon emissions compared to Rule-Based Controllers (RBC) in energy community management. PPO demonstrated highest training stability due to its clipped surrogate objective preventing destructive large policy updates.

Deep Q-Network (DQN) and Hierarchical Approaches: A 2024 study proposed Hierarchical Deep Q-network (HDQN) for microgrid energy dispatching, employing two-layer agent optimization for processing discrete and continuous variables simultaneously, minimizing system operation costs while considering line losses.

Actor-Critic Methods (A3C, DDPG, TD3): The Memory-library A3C (M-A3C) algorithm demonstrated superior strategy optimization compared to DQN, PPO, and double DQN in microgrid scheduling. Twin Delayed Deep Deterministic Policy Gradient (TD3PG) showed effectiveness for optimal generation control in DC microgrids with high training stability.

3. Multi-Time-Scale Energy Management

Island Microgrid Research (2025) developed multi-time-scale scheduling integrating DRL with discrete wavelet transform to decompose and mitigate power fluctuations:

- Day-ahead stage: Hydrogen production and blending optimization
- First intraday stage: Lithium battery and hydrogen equipment adjustments
- Real-time stage: Supercapacitor deployment for fast fluctuations

This hierarchical approach addressed the gap in previous works that focused on individual factors rather than integrated multi-stage frameworks.

4. Hybrid and IoT-Enabled Systems

AI and IoT Integration (2024-2025):

A 2025 study presented an advanced EMS using IoT and AI technologies, employing Generative Adversarial Networks (GANs) to simulate extreme weather scenarios. Key achievements:

- Battery contribution to peak load: 31.8% (increased from 14.5%)
- Daily cost savings: 17% compared to traditional systems
- Average peak load dependence reduction: 7.3 kW
- Four operating modes: Dynamic/Economic with/without load shedding

Joshi et al. (2023) provided a comprehensive survey of AI and ML techniques for microgrid EMS, analyzing methods for centralized, decentralized, and distributed microgrids separately. The review covered ANNs, federated learning, LSTMs, RNNs, and reinforcement learning for objectives including economic dispatch and optimization.

5. Multi-Microgrid Coalitions and Energy Trading

Mahmoudabadi et al. (2025) proposed an energy management framework for optimum scheduling across

multiple interconnected microgrids using machine learning for renewable energy prediction. The two-step approach involved:

- Step 1: Individual microgrid resource scheduling
- Step 2: Distribution system operator decision-making for inter-microgrid power transfers
- Integration of coalition strategies for local and global energy trading

6. MATLAB/Simulink Implementation Studies

Recent Simulation Frameworks:

Research published in 2024-2025 demonstrated MATLAB/Simulink implementations for hybrid renewable energy systems including:

- PV arrays with MPPT algorithms (Perturb & Observe, Incremental Conductance)
- Wind turbines with mechanical and power conversion systems
- Battery management systems with intelligent controllers
- Fuzzy logic-based EMS for power balance management
- Grid synchronization and islanding detection

MathWorks official resources provide comprehensive microgrid models including:

- IEEE 39-bus system examples
- Microgrid resynchronization with BESS
- Grid-forming converter implementations
- Droop control for islanded operation
- HIL (Hardware-in-the-Loop) testing capabilities

7. Comparative Performance Metrics

Recent studies report the following benchmark performances:

Forecasting Accuracy:

- LSTM for load: MAE 4.8%, superior to MLP
- SVR for solar: MAE 0.547, RMSE 1.415
- Random Forest for energy prediction: Strong performance in multi-feature scenarios

Optimization Results:

- DRL-based systems: 20-40% grid dependency reduction
- Cost savings: 15-23% compared to conventional methods
- Renewable utilization: 85-90% in optimized systems

Research Gaps and Opportunities

Despite significant progress, the following gaps remain:

1. Limited integration of multiple ML algorithms in unified frameworks
2. Insufficient validation with real-world, high-resolution data (15-min intervals)
3. Lack of comprehensive MATLAB/Simulink models combining forecasting and DRL
4. Need for adaptive systems handling both normal and contingency scenarios
5. Limited studies on practical deployment with computational constraints

This project addresses these gaps by developing an integrated intelligent system validated through realistic simulations.

SCOPE AND GAPS

Scope of the Project

In-Scope Components:

- 1. Renewable Energy Forecasting Module**
 - Solar PV power generation forecasting using LSTM and SVR
 - Wind power generation forecasting
 - Short-term (15-min to 1-hour) and day-ahead prediction
 - Weather data integration and preprocessing
- 2. Load Demand Forecasting**
 - Historical consumption pattern analysis
 - LSTM-based load prediction
 - Seasonal and time-of-day variation handling
- 3. Energy Storage Management**
 - Battery State-of-Charge (SoC) prediction using ANN/LSTM
 - Charging/discharging optimization
 - Battery degradation consideration
 - Peak shaving and load leveling strategies
- 4. Intelligent Energy Management System**
 - Deep Reinforcement Learning (PPO/DQN) for real-time dispatch
 - Multi-objective optimization: cost, emissions, reliability
 - Rule-based backup strategies for safety
 - Demand response integration

5. MATLAB/Simulink Implementation

- Hybrid microgrid model (Solar PV + Wind + Battery + Grid)
- Power electronic converters and controllers
- Grid-connected and islanded operation modes
- Performance evaluation under various scenarios

6. Performance Analysis

- Forecasting accuracy metrics (MAE, RMSE, MAPE)
- Economic analysis (cost savings, ROI)
- Energy efficiency metrics (renewable utilization, grid dependency)
- System stability and reliability assessment

Out-of-Scope Elements:

1. Hardware prototyping and physical implementation
2. Advanced hydrogen storage systems
3. Electric vehicle (EV) integration and V2G/V2H
4. Peer-to-peer (P2P) energy trading platforms
5. Blockchain-based transaction systems
6. Real-time cyber-attack detection and mitigation
7. Multi-microgrid coalition and networking
8. Detailed electrochemical battery modeling
9. Long-term degradation studies (>5 years simulation)
10. Regulatory and policy framework analysis

Identified Research Gaps

1. Integration Gap

- Most existing research focuses on either forecasting OR optimization separately
- Limited studies combining multiple ML algorithms in one framework
- Need for unified systems integrating LSTM forecasting + DRL optimization

2. Validation Gap

- Many studies use simplified datasets or synthetic data
- Limited validation with high-resolution real-world data (15-min intervals)
- Need for comprehensive scenario testing (normal, fault, extreme weather)

3. Implementation Gap

- Theoretical ML models without practical MATLAB/Simulink implementation
- Gap between research algorithms and engineering deployment

- Limited open-source models for educational and research purposes

4. Scalability Gap

- Most DRL approaches tested on small-scale systems
- Computational complexity for real-time deployment not addressed
- Need for lightweight models suitable for embedded controllers

5. Multi-Time-Scale Gap

- Lack of integrated approaches for different time horizons (real-time to day-ahead)
- Coordination between fast battery response and slow grid interactions
- Need for hierarchical control architectures

How This Project Addresses the Gaps

1. **Integrated Approach:** Combines LSTM/SVR forecasting with DRL-based optimization in single framework
 2. **Realistic Validation:** Uses actual weather and load data from reliable sources (NASA, NREL)
 3. **Practical Implementation:** Complete MATLAB/Simulink model for engineering validation
 4. **Comparative Analysis:** Evaluates multiple algorithms (LSTM vs SVR, PPO vs DQN) to identify best approaches
 5. **Educational Value:** Creates comprehensive, well-documented model for academic use
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OBJECTIVES

Primary Objective

To design, develop, and validate an Intelligent Smart Microgrid Energy Management System using machine learning techniques that optimizes energy generation, storage, and consumption while maximizing renewable energy utilization and minimizing operational costs.

Specific Objectives

1. Renewable Energy Forecasting

- Develop LSTM-based solar PV power forecasting model achieving $RMSE < 2.0$
- Implement wind power prediction using SVR with $RMSE < 2.5$
- Achieve forecasting accuracy (MAPE) $< 5\%$ for short-term predictions
- Compare performance with traditional methods (ARIMA, Linear Regression)

2. Load Demand Prediction

- Design LSTM network for load forecasting with $MAE < 5\%$
- Capture daily, weekly, and seasonal consumption patterns

- Integrate weather dependency for improved predictions

3. Energy Storage Optimization

- Develop ANN/LSTM model for Battery SoC prediction with >95% accuracy
- Implement intelligent charging/discharging strategies
- Achieve 20-30% improvement in battery utilization efficiency
- Prevent SoC violations through predictive control

4. Intelligent Energy Management

- Implement Deep Reinforcement Learning (PPO or DQN) for real-time dispatch
- Achieve 15-20% reduction in operational costs vs. rule-based systems
- Reduce peak grid imports by 15-20%
- Improve renewable energy utilization to 85-90%
- Minimize grid dependency by 35-40%

5. MATLAB/Simulink Implementation

- Develop comprehensive hybrid microgrid model with:
 - Solar PV array with MPPT controller
 - Wind turbine with power converters
 - Battery Energy Storage System (BESS)
 - Grid connection with synchronization
- Simulate grid-connected and islanded operation modes
- Implement control algorithms and power management strategies

6. Performance Evaluation

- Conduct scenario analysis: normal, peak load, renewable variability
- Generate comparative results: ML vs. conventional methods
- Perform economic analysis: cost savings, payback period
- Validate system stability and reliability metrics

7. Documentation and Knowledge Transfer

- Prepare comprehensive technical documentation
- Create user guide for MATLAB/Simulink model
- Publish findings in conference/journal (if applicable)

Success Criteria

The project will be considered successful if:

- Forecasting models achieve target accuracy metrics (RMSE, MAE, MAPE)
- DRL-based EMS demonstrates 15%+ cost reduction vs. baseline

- MATLAB/Simulink model operates successfully in all test scenarios
- System achieves >85% renewable energy utilization
- Complete documentation enables model replication

METHODOLOGY & SYSTEM ARCHITECTURE

Overall System Architecture



- State: Generation, Demand, SoC, Price
- Actions: Charge, Discharge, Grid trade
- Reward: Cost minimize
- Algorithm: PPO/DQN

BATTERY SoC
PREDICTION
(ANN/LSTM)

DISPATCH
DECISIONS

- Battery Ctrl

- Grid Trade

- Load Shed

MATLAB/SIMULINK MODEL

Solar PV
Wind Turbine

DC Bus / AC Bus

Battery BESS
Grid Connect

PERFORMANCE METRICS

- Cost Savings
- Renewable Utilization
- Forecast Accuracy
- Grid Dependency

Implementation Phases

Phase 1: Data Collection & Preprocessing (Weeks 1-2)

- Obtain weather data from NASA/NREL databases
- Collect load consumption patterns
- Clean and normalize datasets
- Feature engineering and correlation analysis

Phase 2: Forecasting Model Development (Weeks 3-5)

- LSTM network design and training for solar/load
- SVR implementation for wind forecasting
- Model validation and hyperparameter tuning
- Comparison with baseline methods

Phase 3: DRL-based EMS Development (Weeks 6-8)

- Define state space, action space, reward function
- Implement PPO/DQN algorithms
- Training using historical data
- Policy evaluation and refinement

Phase 4: MATLAB/Simulink Implementation (Weeks 9-11)

- Develop microgrid component models
- Integrate ML models via Python-MATLAB interface
- Implement control strategies
- Testing and debugging

Phase 5: Validation & Analysis (Weeks 12-13)

- Scenario testing (normal, peak, fault)
- Performance metric calculation
- Comparative analysis
- Sensitivity analysis

Phase 6: Documentation & Presentation (Week 14)

- Technical report preparation
- Presentation slides
- Demo video creation

- Final submission
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SYSTEM REQUIREMENTS

Software Requirements

Primary Development Tools:

1. MATLAB R2023b or later
 - Simulink
 - Simscape Electrical (Power Systems)
 - Deep Learning Toolbox
 - Reinforcement Learning Toolbox
 - Optimization Toolbox
 - Statistics and Machine Learning Toolbox
2. Python 3.8 or later
 - TensorFlow 2.x or PyTorch 1.x
 - NumPy, Pandas, Scikit-learn
 - Matplotlib, Seaborn (visualization)
 - Stable-Baselines3 (for DRL)
3. Data Sources
 - NASA POWER Data Access Viewer (weather data)
 - NREL NSRDB (National Solar Radiation Database)
 - Public microgrid datasets (Portugal, Rye, etc.)

Optional Tools:

- Jupyter Notebook (for ML model development)
- Git (version control)
- LaTeX (documentation)

Hardware Requirements

Minimum Specifications:

- Processor: Intel Core i5 (8th gen) or equivalent
- RAM: 16 GB
- Storage: 50 GB free space
- GPU: Optional but recommended (NVIDIA with CUDA support)

Recommended Specifications:

- Processor: Intel Core i7/i9 or AMD Ryzen 7/9
- RAM: 32 GB
- Storage: SSD with 100 GB free space
- GPU: NVIDIA RTX 3060 or better (for faster training)

Data Requirements

Input Datasets:

1. Weather Data (1 year minimum):

- Solar irradiance (W/m^2)
- Temperature ($^{\circ}\text{C}$)
- Wind speed (m/s)
- Cloud cover (%)
- Temporal resolution: 15-min or 1-hour

2. Load Data:

- Historical consumption (kWh)
- Peak/off-peak patterns
- Seasonal variations
- Resolution: 15-min to 1-hour

3. System Parameters:

- Solar PV capacity and efficiency
 - Wind turbine specifications
 - Battery capacity and characteristics
 - Grid tariff structure
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PROJECT TIMELINE & MILESTONES

14-Week Implementation Plan

Week	Phase	Tasks	Deliverables	Status
1-2	Data Collection	- Identify and download weather datasets - Collect load consumption data - Data cleaning and preprocessing - Exploratory data analysis	- Clean datasets ready - Data analysis report	Not Started
3	Forecasting - Solar	- LSTM architecture design - Training for solar PV prediction - Hyperparameter tuning	- Trained LSTM model - Initial accuracy metrics	Not Started
4	Forecasting - Wind & Load	- SVR implementation for wind - LSTM for load forecasting - Model validation	- Complete forecasting suite - Comparative results	Not Started
5	Forecasting Evaluation	- Benchmark against baselines - Error analysis - Model refinement	- Forecasting report - Accuracy benchmarks	Not Started
6	DRL - Design	- Define MDP (states, actions, rewards) - Select RL algorithm (PPO/DQN) - Design network architecture	- DRL design document - Reward function spec	Not Started
7	DRL - Training	- Implement RL algorithm - Training with historical data - Policy evaluation	- Trained RL agent - Learning curves	Not Started
8	DRL - Optimization	- Fine-tuning and testing - Compare with rule-based - Sensitivity analysis	- Optimized RL policy - Performance comparison	Not Started
9	Simulink - Components	- Solar PV model with MPPT - Wind turbine model - Battery BESS model	- Individual component models - Unit test results	Not Started
10	Simulink - Integration	- DC/AC bus configuration - Grid connection model - Controller implementation	- Integrated microgrid model - Preliminary tests	Not Started
11	Simulink - ML Integration	- Python-MATLAB interface - ML model deployment	- Complete EMS system - Interface testing	Not Started

Week	Phase	Tasks	Deliverables	Status
		• Control strategy implementation		
12	Testing & Validation	• Normal operation scenarios	• Test results report	Not
		• Peak load testing	• Bug fixes	Started
		• Fault condition analysis		
13	Performance Analysis	• Calculate metrics	• Performance report	Not
		• Economic analysis	• Analysis charts	Started
		• Comparative evaluation		
14	Documentation	• Technical report	• Final report	Not
		• Presentation preparation	• Presentation	Started
		• Demo video	• Demo	

Key Milestones

Milestone 1 (End of Week 2): Data Ready

- All datasets collected and preprocessed
- Feature engineering completed

Milestone 2 (End of Week 5): ML Models Complete

- Forecasting models trained and validated
- Accuracy targets achieved

Milestone 3 (End of Week 8): DRL Optimization Done

- RL agent trained and optimized
- Performance superior to baseline

Milestone 4 (End of Week 11): Simulink Model Complete

- Full microgrid model operational
- ML integration successful

Milestone 5 (End of Week 13): Validation Complete

- All scenarios tested
- Performance metrics calculated

Milestone 6 (End of Week 14): Project Complete

- Documentation finalized
- Presentation ready

EXPECTED OUTCOMES & PERFORMANCE METRICS

Forecasting Accuracy Targets

Solar PV Forecasting:

- RMSE: < 2.0 kW (for 100 kW system)
- MAE: < 0.6 kW
- MAPE: < 5%
- R² Score: > 0.95

Wind Power Forecasting:

- RMSE: < 2.5 kW (for 50 kW system)
- MAE: < 0.9 kW
- MAPE: < 6%
- R² Score: > 0.93

Load Demand Forecasting:

- RMSE: < 3.0 kW
- MAE: < 1.0 kW
- MAPE: < 5%

Energy Management Performance

Operational Efficiency:

- Renewable Energy Utilization: 85-90%
- Grid Dependency Reduction: 35-40%
- Peak Grid Import Reduction: 15-20%
- Energy Self-Sufficiency: > 85%

Economic Metrics:

- Operational Cost Reduction: 20-25% vs. baseline
- Daily Cost Savings: 15-20%
- Battery Utilization Improvement: 25-30%
- Payback Period: < 5 years (in real scenario)

Battery Management:

- SoC Prediction Accuracy: > 95%
- Peak Load Battery Contribution: 25-35%
- SoC Violations: < 2% of operation time
- Charging/Discharging Efficiency: > 90%

System Reliability:

- Voltage Deviation: < 5% of nominal
- Frequency Deviation: < 0.5 Hz (if applicable)
- Unmet Load: < 1% annually
- System Availability: > 99%

Comparative Performance

vs. Rule-Based System:

- Cost: 20-25% reduction
- Renewable Utilization: +10-15%
- Peak Demand: -15-20%

vs. Traditional Forecasting (ARIMA):

- RMSE Improvement: 30-40%
- MAPE Improvement: 25-35%

vs. Conventional Optimization (MILP):

- Computation Time: 10-100x faster
- Adaptability: Real-time vs. offline
- Scalability: Better for large systems

REFERENCES (SAMPLE - TO BE EXPANDED)

Key Research Papers (2023-2025)

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7. ScienceDirect (2025). "A review of machine learning and IoT-based energy management systems for AC microgrids." *Renewable and Sustainable Energy Reviews*.
8. Springer (2024). "An intelligent model for efficient load forecasting and sustainable energy management in sustainable microgrids." *Discover Sustainability*.

MATLAB/Simulink Resources

9. MathWorks. "Microgrid, Smart Grid, and Charging Infrastructure - MATLAB & Simulink." <https://www.mathworks.com/solutions/electrification/microgrid-smart-grid-charging-infrastructure.html>
10. MathWorks. "What Is Microgrid Control?" <https://www.mathworks.com/discovery/microgrid-control.html>
11. MATLAB Central File Exchange. "Microgrid Hybrid PV/Wind/Battery Management System." <https://www.mathworks.com/matlabcentral/fileexchange/60550>

Data Sources

12. NASA POWER Project. "Prediction Of Worldwide Energy Resources." <https://power.larc.nasa.gov/>
 13. NREL. "National Solar Radiation Database (NSRDB)." <https://nsrdb.nrel.gov/>
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PRELIMINARY WORK DONE (IF ANY)

Status: Project Initiation Phase

Currently completed:

- Literature review of 15+ recent papers (2023-2025)
- Identification of research gaps
- Selection of ML algorithms (LSTM, SVR, PPO/DQN)
- Project scope definition
- Timeline planning

To be initiated:

- Data collection
- Software setup and environment configuration

- Model development
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CONCLUSION

This project aims to develop a comprehensive Intelligent Smart Microgrid Energy Management System that addresses critical challenges in renewable energy integration through advanced machine learning techniques. By combining LSTM/SVR forecasting with deep reinforcement learning optimization, implemented in a realistic MATLAB/Simulink environment, the system promises significant improvements in operational efficiency, cost reduction, and renewable energy utilization.

The project is well-positioned to contribute to both academic research and practical engineering applications, providing a validated framework that can be extended for real-world microgrid deployments. Expected outcomes include 85-90% renewable energy utilization, 20-25% cost reduction, and accurate forecasting with $MAPE < 5\%$, demonstrating the transformative potential of AI-driven energy management systems.

Document prepared for B.Tech Final Year Project

Department of Electrical and Electronics Engineering

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